

Exercises 06

Implementing Selection on Multiple Attributes (N-d)

[\[Show with no answers\]](#) [\[Show with all answers\]](#)

1. Consider a file of $n=50000$ tuples allocated across $b=1024$ pages using a multi-attribute hash function giving $d=10$ hash bits. The tuples in this file have four fields $R(w,x,y,z)$ and a choice vector that allocates hash bits to fields as follows: $d_w=5$, $d_x=2$, $d_y=3$, $d_z=0$. Assuming that there are no overflow pages, compute how many pages each of the following queries would need to access:

a. `select * from R where w=5432 and x=3`

[\[show answer\]](#)

b. `select * from R where w=4523 and x=9 and y=12`

[\[show answer\]](#)

c. `select * from R where x=3`

[\[show answer\]](#)

d. `select * from R where z=3`

[\[show answer\]](#)

e. `select * from R where w=9876 and x>5`

[\[show answer\]](#)

2. Consider a file of $r=819,200$ `Part` records ($C=100$):

```
CREATE TABLE Parts (  
    id#      number(10) primary key,  
    name     varchar(10),  
    colour   varchar(5) check value in ('red','blue','green'),  
    onhand   integer  
);
```



Used only via the following kinds of *pmr* queries:

Query Type	p_Q
$\langle \text{id\#, ?}, ?, ? \rangle$	0.25
$\langle ?, \text{name}, \text{colour}, ? \rangle$	0.50
$\langle ?, ?, \text{colour}, ? \rangle$	0.25

Give and justify values for d and the d_i s and suggest a suitable choice vector.

[\[show answer\]](#)

3. Consider the student relation:

```
Student(id:integer, name:string, address:string,
        age:integer, course:string, gpa:real);
```

with the following characteristics: $r = 40,000$, $B = 1024$, $C = 20$

If the relation is accessed via a superimposed codeword signature file with false match probability $p_F = 10^{-4}$, compute the costs of answering the query:

```
select * from Student where course='BSc' and age=20;
```

for the following file organisations:

- record signatures
- block signatures
- bit-sliced block signatures

Use the following to compute signature properties:

$$k = \frac{1}{\log_e 2} \cdot \log_e \left(\frac{1}{p_F} \right) \quad m = \left(\frac{1}{\log_e 2} \right)^2 \cdot n \cdot \log_e \left(\frac{1}{p_F} \right)$$

[\[show answer\]](#)

4. Consider a multi-attribute hashed relation with the following properties:

- schema $R(a, b, c)$, where all attributes are integers
- a file with pages $b=2$, depth $d=1$, split pointer $sp=0$, records/page $C=2$
- a split occurs after every 3 insertions
- an initially empty overflow file
- choice vector = $\langle (1,0), (2,0), (3,0), (1,1), (1,2), (2,1), (2,2), (3,1), \dots \rangle$
- the hash value for each attribute is simply the binary version of the value
(e.g. $\text{hash}(0) = \dots 0000$, $\text{hash}(1) = \dots 0001$, $\text{hash}(4) = \dots 0100$, $\text{hash}(11) = \dots 1011$, etc.)



Show the state of the data and overflow files after the insertion of the following tuples (in the order given):

(3,4,5)	(2,4,6)	(2,3,4)	(3,5,6)	(4,3,2)	(2,6,5)	(4,5,6)	(
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[\[show answer\]](#)

